

EXPERIMENT 7

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Experiment 7 – Implement Router as DHCP server that can serve multiple VLAN's.

Aim

To implement Virtual Local Area Networks (VLANs) and VLAN trunking on network switches, and to enable communication between different VLANs using appropriate trunking and routing techniques.

Objective

- To understand the concept and purpose of VLANs in a computer network.
- To create and configure multiple VLANs on a network switch.
- To assign switch ports to specific VLANs.
- To configure trunk links for carrying traffic from multiple VLANs.
- To understand VLAN trunking using the IEEE 802.1Q protocol.
- To verify VLAN connectivity and communication using suitable network commands.

Theory

VLAN (Virtual Local Area Network)

A VLAN is a logical network created within a physical switch. It allows devices connected to the same physical switch to be divided into separate logical networks.

For example, **VLAN 10** can be assigned to Students and **VLAN 20** to Faculty. Although both groups may use the same physical switch, VLANs logically separate their network traffic.

Need for VLANs

VLANs are used to divide a network into smaller logical segments. They provide the following benefits:

- Network segmentation.
- Improved network security.
- Reduction of broadcast traffic.
- Better network management.
- Logical grouping of users and devices.

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Inter-VLAN Routing

Different VLANs are separate Layer 2 networks. A Layer 2 switch cannot normally route traffic between different VLANs. Communication between VLANs therefore requires a Layer 3 device, such as a router or Layer 3 switch.

This process is called **Inter-VLAN Routing**.

Router-on-a-Stick

Router-on-a-Stick is a method of performing Inter-VLAN Routing using a single physical router interface divided into multiple logical subinterfaces.

For example:

- **G0/0.10** for VLAN 10.
- **G0/0.20** for VLAN 20.

This allows one physical router interface to handle traffic from multiple VLANs.

Subinterface

A subinterface is a logical interface created from a physical router interface. It allows one physical interface to handle traffic belonging to multiple VLANs.

VLAN Tagging and 802.1Q

When traffic from multiple VLANs travels through a trunk link, VLAN tagging identifies the VLAN to which each Ethernet frame belongs.

IEEE 802.1Q is a standard used to tag VLAN traffic over trunk links.

For example:

`encapsulation dot1Q 10`

associates a router subinterface with **VLAN 10**.

Default Gateway

A default gateway is the device to which a host sends packets when the destination is outside its local network.

In this experiment, the router subinterface acts as the default gateway for its respective VLAN.

For example:

- **192.168.10.1** → Gateway for VLAN 10.
- **192.168.20.1** → Gateway for VLAN 20.

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DHCP

DHCP stands for **Dynamic Host Configuration Protocol**. It automatically provides network configuration information to client devices, including:

- IP address.
- Subnet mask.
- Default gateway.
- DNS server information.
- Lease information.

DHCP reduces the need to manually configure each computer.

Router as DHCP Server

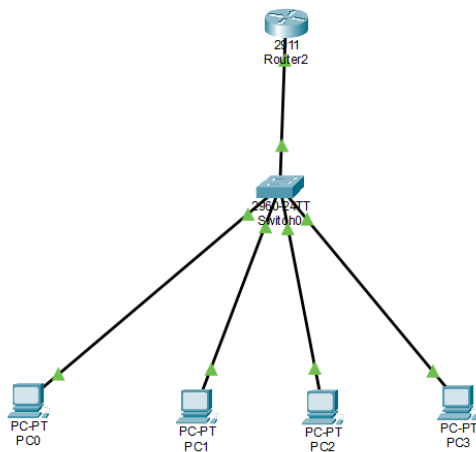
A Cisco router can be configured as a DHCP server. In this experiment, a separate DHCP server is not required because the router provides both **DHCP services** and **Inter-VLAN Routing** for the different VLANs.

Practical Steps

1. Open **Cisco Packet Tracer** and place one 2911 router, one 2960 switch, and four PCs.
2. Rename the devices as **R1, S1, PC1, PC2, PC3, and PC4**, and connect them using Copper Straight-Through cables.
3. Create **VLAN 10** and **VLAN 20** on the switch and assign the appropriate PC ports as access ports.
4. Configure the switch port connected to the router as a **trunk port**.
5. Verify the VLAN and trunk configuration using:
 - show vlan brief
 - show interfaces trunk
6. Configure the router interface and create subinterfaces **G0/0.10** and **G0/0.20** using **802.1Q encapsulation**.
7. Assign **192.168.10.1** as the gateway for VLAN 10 and **192.168.20.1** as the gateway for VLAN 20.
8. Configure the router as a **DHCP server** with separate pools for VLAN 10 and VLAN 20, and exclude the gateway addresses.
9. Configure all PCs to obtain their IP addresses automatically using **DHCP**.

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10. Verify that VLAN 10 PCs receive **192.168.10.x** addresses and VLAN 20 PCs receive **192.168.20.x** addresses.
11. Test connectivity within the same VLAN and between different VLANs using the **Ping** command.
12. Verify that the router is successfully performing **Inter-VLAN Routing** and providing IP addresses through DHCP.



The screenshot shows the Packet Tracer configuration window for a packet capture on the PFE100 router. The configuration is set to capture on the 'PC0' source and 'PC2' destination. The capture type is 'ICMP' and the capture color is 'Red'. The capture time is set to '0.000' seconds. The capture is periodic with a duration of 'N' and a number of captures of '1'. The capture is named 'PDU List Window'.

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Learning Outcome

1. Understand the concept of VLANs and their role in logically separating devices into different broadcast domains.
2. Develop practical skills in creating VLANs and assigning switch ports to appropriate VLANs using Cisco Packet Tracer.
3. Understand the difference between **access ports** and **trunk ports**, and configure a trunk connection between a switch and router.
4. Understand **Inter-VLAN Routing** and the **Router-on-a-Stick** method using router subinterfaces.
5. Configure router subinterfaces for multiple VLANs and assign appropriate default gateway addresses.
6. Configure a Cisco router as a **DHCP server** and create separate DHCP pools for different VLANs.
7. Automatically assign IP addresses, subnet masks, and default gateways to PCs using DHCP.
8. Verify VLAN, trunk, DHCP, and routing configurations and test communication between devices belonging to different VLANs.